Restaurant Data Warehouse

Star Schema Design (SQL Server)

1. Overview

This document is the Star Schema version of the restaurant data warehouse. Every dimension joins directly to the fact tables, with no outer lookup tables: hierarchies such as category, position, unit and month are flattened into their parent dimensions. The restaurant has a single branch, so there is no branch dimension.

Fact tables and grain

- FactOrderLine: one menu item on one order
- FactPayment: one payment
- FactPurchaseLine: one item on one purchase order
- FactInventoryTransaction: one stock movement

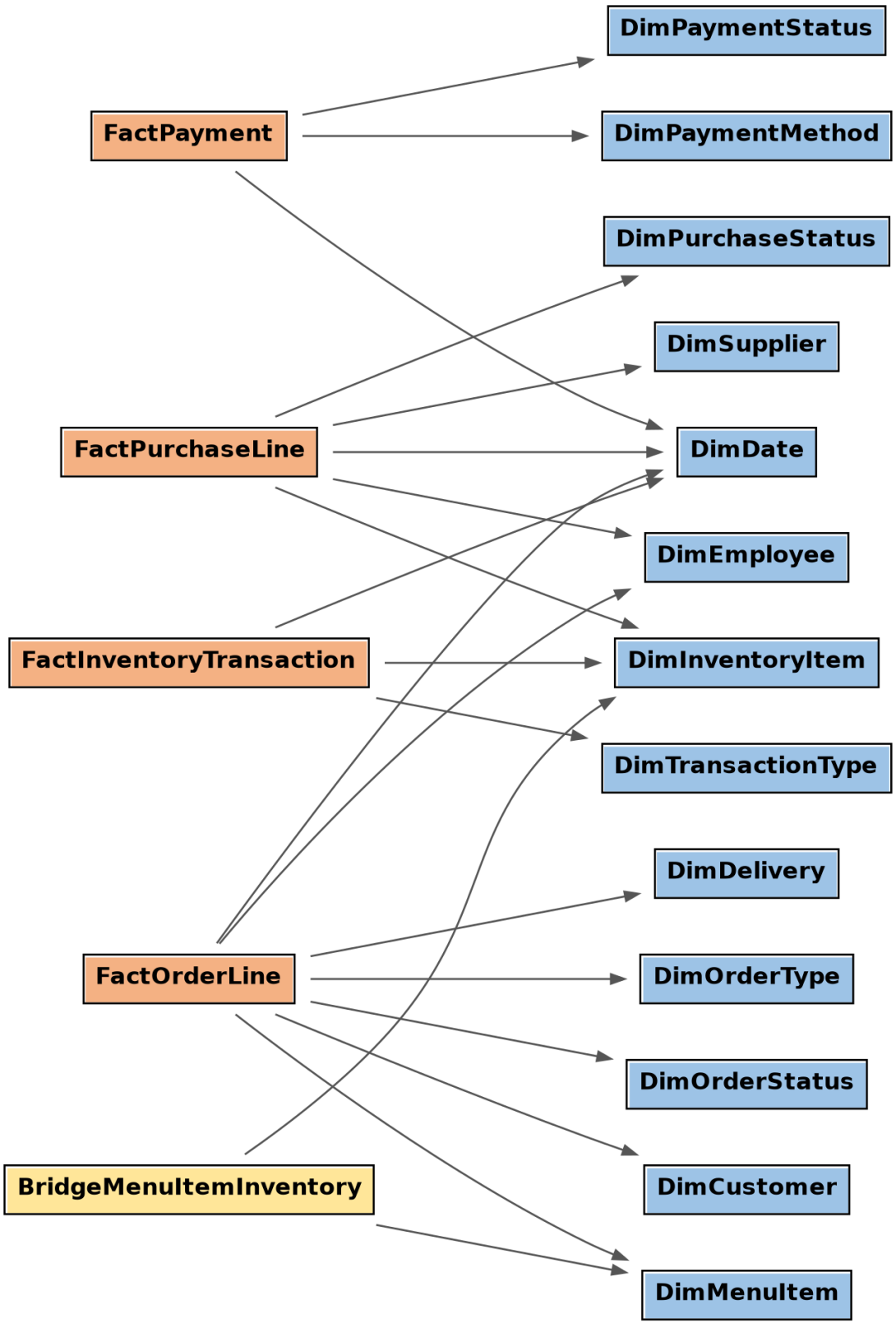
Key legend

PK = surrogate primary key, FK = foreign key, BK = business key from the source system, DD = degenerate dimension (kept on the fact table), Measure = numeric measure.

2. Star Schema Diagrams

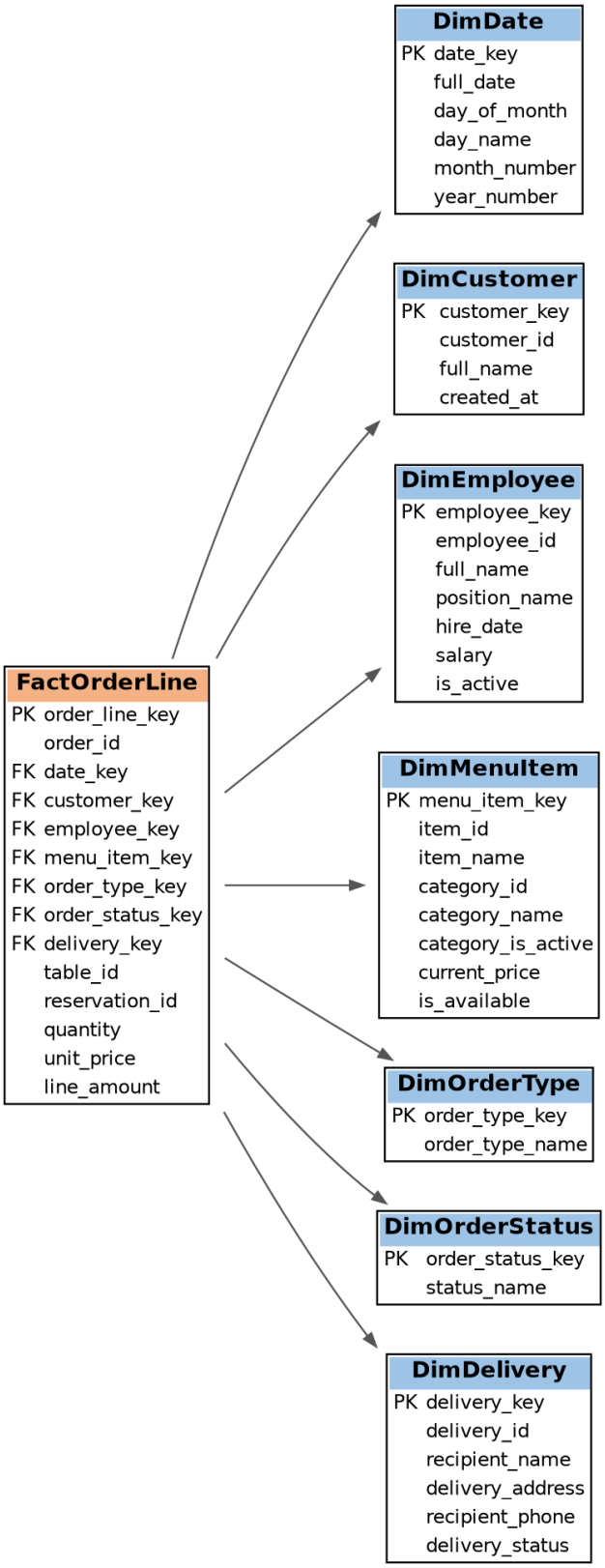
Orange = fact, blue = dimension, yellow = bridge. Arrows point from the referencing table to the referenced table.

Overview (table names only)

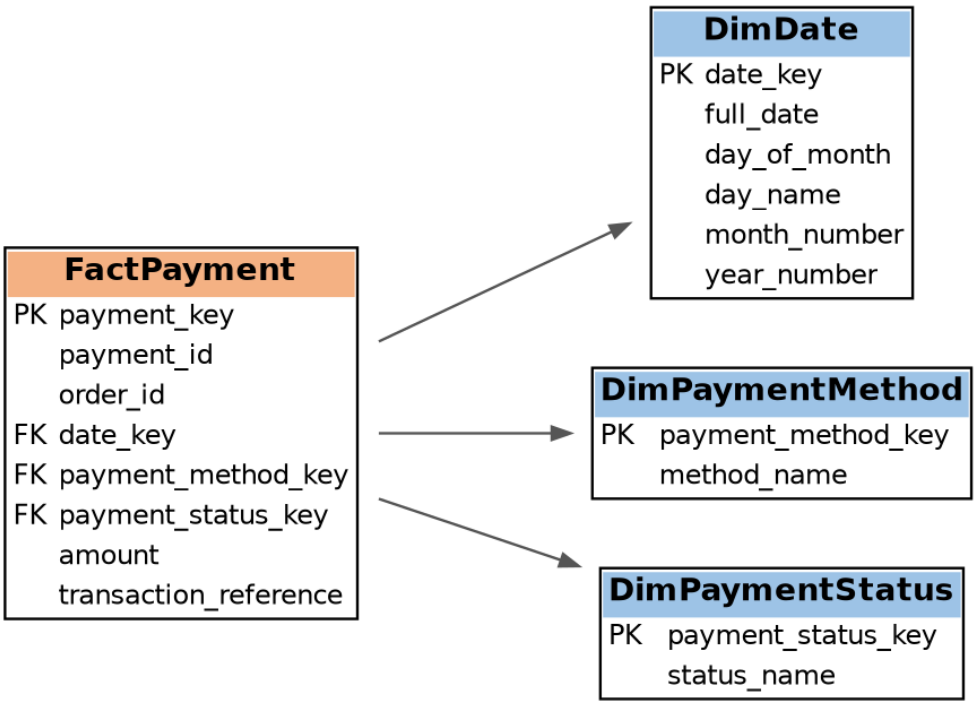


2.1 Star Views Per Fact Table

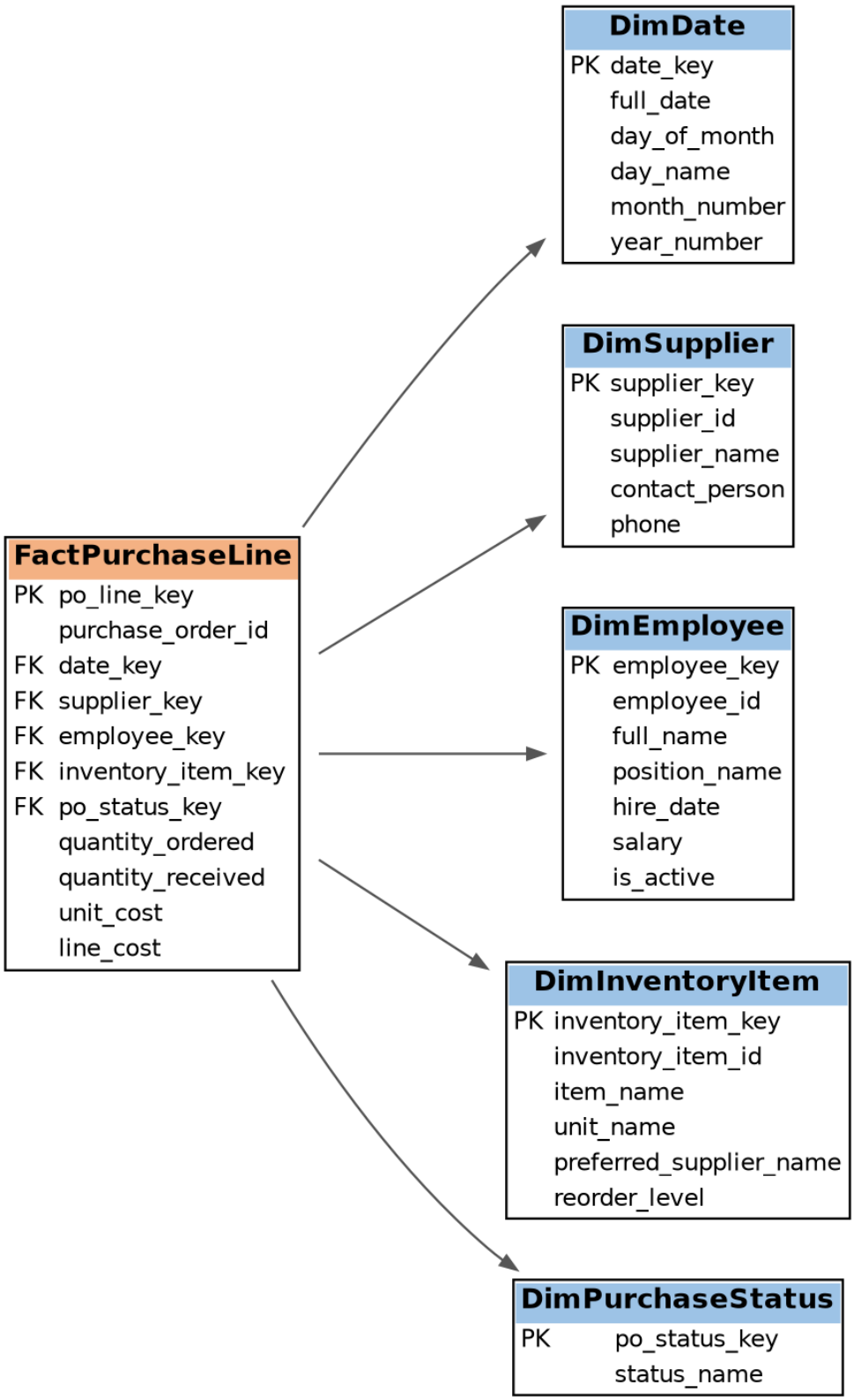
FactOrderLine star



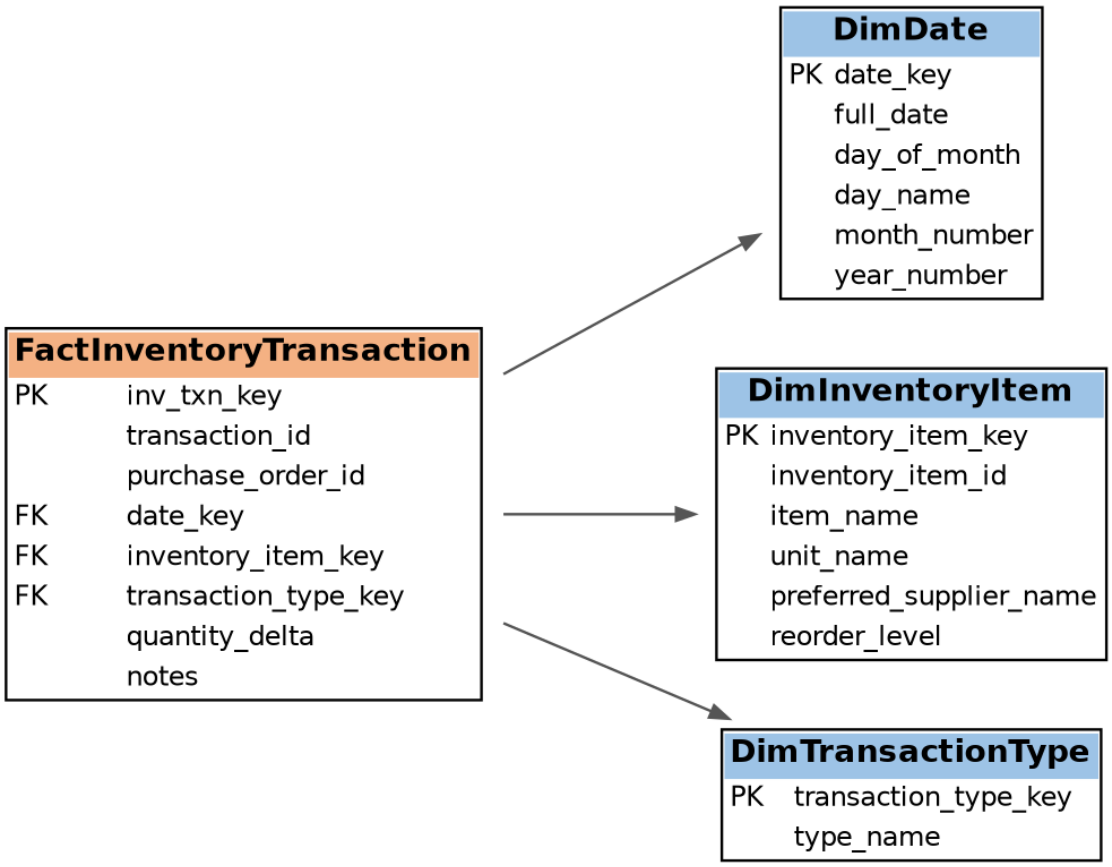
FactPayment star



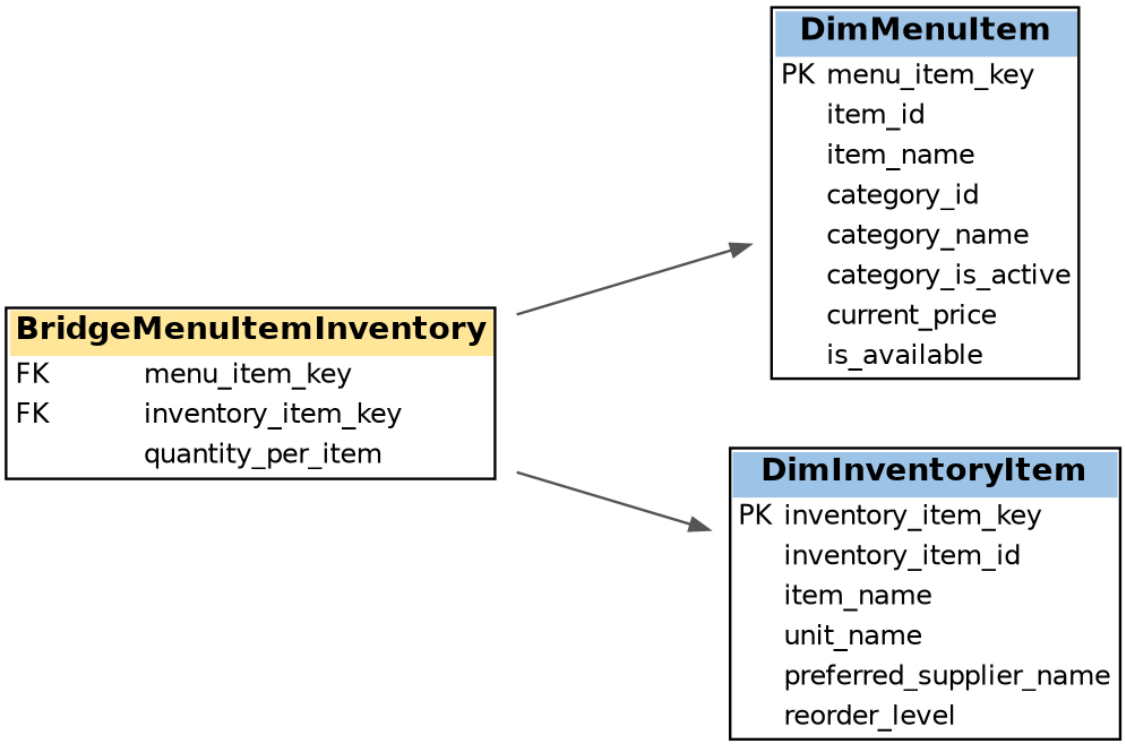
FactPurchaseLine star



FactInventoryTransaction star



Bridge table (recipe)



3. Changes and Design Decisions

Requested changes

- DimCustomer and DimEmployee: phone and email removed.
- DimDate: is_weekend removed.
- DimMonth: month_name and quarter_number removed. The remaining month_number and year_number are merged into DimDate, since a star schema has no separate month table.
- Status split out: DimOrderStatus is its own dimension, separate from DimOrderType, and DimPaymentStatus is its own dimension, separate from DimPaymentMethod. Both connect directly to the fact table.
- Single branch: DimBranch, branch_key on the facts, and BridgeBranchMenuItem are removed. Menu pricing uses current_price on DimMenuItem; the old branch_price column is not carried over.

Star schema flattening

- DimMenuItem now contains the category columns (DimCategory removed).
- DimEmployee now contains position_name (DimPosition removed).
- DimInventoryItem now contains unit_name and preferred_supplier_name (DimUnit removed). DimSupplier stays as its own dimension for purchasing.
- DimDelivery now contains delivery_status (DimDeliveryStatus removed).

Carried over from the previous design

- Order header and lines are merged into FactOrderLine; the header total_amount is not repeated on each line.
- BridgeMenuItemInventory is the recipe bridge between menu items and stock items. Its quantity column is renamed quantity_per_item as an assumption and should be confirmed.
- table_id and reservation_id stay as degenerate dimensions because the Table and Reservation tables are not in the source.
- quantity_on_hand is a changing snapshot, so it is not stored in a dimension. A periodic snapshot fact can be added if needed.

4. Fact Tables

FactOrderLine

Column	Data type	Key	Notes
order_line_key	BIGINT	PK	
order_id	INT	DD	Degenerate dimension
date_key	INT	FK	
customer_key	INT	FK	
employee_key	INT	FK	
menu_item_key	INT	FK	
order_type_key	INT	FK	
order_status_key	INT	FK	
delivery_key	BIGINT	FK	Nullable (delivery orders only)
table_id	INT	DD	Degenerate dimension, nullable
reservation_id	INT	DD	Degenerate dimension, nullable
quantity	INT	Measure	
unit_price	DECIMAL(12,2)	Measure	
line_amount	DECIMAL(12,2)	Measure	

FactPayment

Column	Data type	Key	Notes
payment_key	BIGINT	PK	
payment_id	INT	DD	Degenerate dimension
order_id	BIGINT	DD	Degenerate dimension
date_key	INT	FK	
payment_method_key	INT	FK	
payment_status_key	INT	FK	
amount	DECIMAL(10,2)	Measure	
transaction_reference	VARCHAR(100)		

FactPurchaseLine

Column	Data type	Key	Notes
po_line_key	BIGINT	PK	
purchase_order_id	BIGINT	DD	Degenerate dimension
date_key	INT	FK	
supplier_key	INT	FK	
employee_key	INT	FK	
inventory_item_key	INT	FK	
po_status_key	INT	FK	

Column	Data type	Key	Notes
quantity_ordered	DECIMAL(12,3)	Measure	
quantity_received	DECIMAL(12,3)	Measure	
unit_cost	DECIMAL(12,2)	Measure	
line_cost	computed	Measure	quantity_received x unit_cost

FactInventoryTransaction

Column	Data type	Key	Notes
inv_txn_key	BIGINT	PK	
transaction_id	BIGINT	DD	Degenerate dimension
purchase_order_id	BIGINT	DD	Degenerate dimension, nullable
date_key	INT	FK	
inventory_item_key	INT	FK	
transaction_type_key	INT	FK	
quantity_delta	DECIMAL(12,3)	Measure	
notes	VARCHAR(250)		

5. Dimension Tables

DimDate

Column	Data type	Key	Notes
date_key	INT	PK	yyyymmdd
full_date	DATE		
day_of_month	TINYINT		
day_name	VARCHAR(10)		
month_number	TINYINT		Merged from DimMonth
year_number	SMALLINT		Merged from DimMonth

DimCustomer

Column	Data type	Key	Notes
customer_key	INT	PK	
customer_id	INT	BK	
full_name	NVARCHAR(100)		
created_at	DATETIME		

DimEmployee

Column	Data type	Key	Notes
employee_key	INT	PK	
employee_id	INT	BK	
full_name	NVARCHAR(100)		
position_name	NVARCHAR(50)		Flattened from DimPosition
hire_date	DATE		
salary	DECIMAL(10,2)		
is_active	BIT		

DimMenuItem

Column	Data type	Key	Notes
menu_item_key	INT	PK	
item_id	INT	BK	
item_name	VARCHAR(100)		
category_id	BIGINT		Flattened from DimCategory
category_name	NVARCHAR(80)		Flattened from DimCategory
category_is_active	BIT		Flattened from DimCategory
current_price	DECIMAL(12,2)		
is_available	BIT		

DimSupplier

Column	Data type	Key	Notes
supplier_key	INT	PK	
supplier_id	BIGINT	BK	
supplier_name	VARCHAR(100)		
contact_person	VARCHAR(100)		
phone	VARCHAR(25)		

DimInventoryItem

Column	Data type	Key	Notes
inventory_item_key	INT	PK	
inventory_item_id	BIGINT	BK	
item_name	VARCHAR(100)		
unit_name	VARCHAR(20)		Flattened from DimUnit
preferred_supplier_name	VARCHAR(100)		Flattened from supplier; nullable
reorder_level	DECIMAL(12,3)		

DimDelivery

Column	Data type	Key	Notes
delivery_key	BIGINT	PK	
delivery_id	BIGINT	BK	
recipient_name	VARCHAR(100)		
delivery_address	VARCHAR(255)		
recipient_phone	VARCHAR(25)		
delivery_status	VARCHAR(20)		Flattened from DimDeliveryStatus

DimOrderType

Column	Data type	Key	Notes
order_type_key	INT	PK	
order_type_name	VARCHAR(20)		

DimOrderStatus

Column	Data type	Key	Notes
order_status_key	INT	PK	
status_name	VARCHAR(20)		

DimPaymentMethod

Column	Data type	Key	Notes
payment_method_key	INT	PK	
method_name	VARCHAR(20)		

DimPaymentStatus

Column	Data type	Key	Notes
payment_status_key	INT	PK	
status_name	VARCHAR(20)		

DimPurchaseStatus

Column	Data type	Key	Notes
po_status_key	INT	PK	
status_name	VARCHAR(20)		

DimTransactionType

Column	Data type	Key	Notes
transaction_type_key	INT	PK	
type_name	VARCHAR(20)		

6. Bridge Table

BridgeMenuItemInventory

Column	Data type	Key	Notes
menu_item_key	INT	FK	
inventory_item_key	INT	FK	
quantity_per_item	DECIMAL(12,3)	Measure	Renamed from quantity_in_stock - confirm meaning